BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

File Handling

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is a parameter?

2. How does phishing trick people?

A) Tries every possible password
B) Floods a server with traffic
C) Repairs damaged files
D) Pretends to be trusted

3. "Fibre carries electrical signals through copper wire." Correct this. Explain why.

4. Desktops and moving tablets need network links. "Wi-Fi needs a cable from each tablet to the switch." Correct this. Which links suit these devices?

Today's objectives

Read, write, open and close text files. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

Open file
open(filename) returns a handle to an existing text file.

Read line
handle.readLine() obtains the next line of text from an opened file.

Write line
handle.writeLine(text) adds a line to an opened file.

Close file
handle.close() releases the file when finished.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Open file	open(filename) returns a handle to an existing text file. In the daily attendance file, identify where this technique applies.
Read line	handle.readLine() obtains the next line of text from an opened file. In the revision notes file, identify where this technique applies.
Write line	handle.writeLine(text) adds a line to an opened file. In the library loans file, identify where this technique applies.
Close file	handle.close() releases the file when finished. In the temperature log, identify where this technique applies.

I Do - One worked example

A daily attendance file uses open file and read line. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Open file [1]; open(filename) returns a handle to an existing text file. [1].

3. Read line [1]; handle.readLine() obtains the next line of text from an opened file. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A revision notes file needs open file. Explain the purpose. [2 marks]

2. Explain read line for a library loans file. [2 marks]

3. When programming a temperature log, explain why write line and close file matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the statement that appends "Backup complete" to an existing text file.

```
myFile = open("status.txt")
// write a line
myFile.close()
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain read line and write line in a sport-result text file. [4 marks]

Q4. A programmer makes a teacher diary file. Explain the roles of open file, write line and close file. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. open(filename) returns a handle to an existing text file. ☐ T ☐ F
2. open(filename) returns a handle to an existing text file ☐ T ☐ F
3. handle.readLine() obtains the next line of text from an opened file ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A named input received by a subprogram when called with an argument. ☐

Q2 D - Pretends to be trusted ☐

Q3 Fibre uses light. It supports fast data transfer over long distances and resists electromagnetic interference. Installation can cost more. ☐

Q4 Ethernet uses cables. Wi-Fi uses radio, so tablets can move around, but interference can affect it. Choose links to suit the distance and data rate needed. ☐

Code First - Model

```
myFile = open("status.txt")
myFile.writeLine("Backup complete")
myFile.close()
```

Output: No screen output; "Backup complete" appended to status.txt

We Do - Check the model

1. Open file [1]; open(filename) returns a handle to an existing text file. [1]. ☐

2. Read line [1]; handle.readLine() obtains the next line of text from an opened file. [1]. ☐

3. Write line [1]; handle.writeLine(text) adds a line to an opened file. [1]. Close file [1]; handle.close() releases the file when finished. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Open file [1]; Read line [1]. ☐

Q3 Read line [1]; handle.readLine() obtains the next line of text from an opened file. [1]. Write line [1]; handle.writeLine(text) adds a line to an opened file. [1]. ☐

Q4 Open file [1]; open(filename) returns a handle to an existing text file. [1]. Write line [1]; handle.writeLine(text) adds a line to an opened file. [1]. Close file [1]; handle.close() releases the file when finished. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: